

Framingham Data Revisited: Isolating the Causal Effect of Education on Stroke Risk

In 1948, an extended prospective cohort study on the etiology of cardiovascular disease, known as the Framingham Study, was launched and continues to this day. The aim of the study is to monitor the development of cardiovascular diseases and to determine which factors influence the risk of developing them over time. It provided many insights that now form the basis for lifestyle recommendations, medications and cardiac surveillance screening (Mahmood et al., 2014).

Participants in the study were followed for 24 years, extensively recording their demographics, occupations and health status. Previous studies have analysed this data aiming to find correlations between risk factors and the development of cardiovascular diseases. For example, Freund et al. (1993) reported on the health risk of smoking, and Garrison et al. (1993) already reported a correlation between educational attainment and the risk of developing cardiovascular diseases.

A fundamental problem in these correlation studies is their translation to causality. Often, only a relationship between two events was found and analytically proven, but not the actual causation of event B by event A. Another common problem is the potential unaccounted influences of other events on the outcome (cardiovascular diseases). These confounding factors can create misleading results in a standard correlation analysis. By controlling for these factors, the data analysis will yield valid conclusions on the correlation, and a valid estimation of the causal effect of event A on event B.

Garrison et al. (1993) is supported in the shown inverse association between cardiovascular diseases and the level of education. However, these associations might be misleading, as there might be these confounding factors strengthening or weakening the statistical correlation. For example: people with a lower education have a higher probability to be a smoker (Latvala et al. 2014), and smokers have a higher risk of developing cardiovascular diseases (Freund et al. (1993). It remains unclear what the actual effect of education on the development of these diseases is.

In this study we aim to analytically estimate a causal effect of education on the development of stroke. Using potential outcome notation, where Y represents stroke outcome (1=Stroke, 0=No Stroke) and X represents the education level (1=College or higher, 0=High school or lower), our target estimand is the Average Treatment Effect (ATE) defined as the risk difference $E[Y(X=0)] - E[Y(X=1)]$. We interpret a positive ATE as lower risk for higher education. We expect that people who have attained higher education have a reduced risk of developing a stroke. By analysing a subset of the Framingham Study data, we will first identify factors involved in and potentially influencing the correlation between education and the development of stroke. Second, we will perform a propensity weighting analysis to control for the confounding factors. Subsequently, we will estimate the causal effect of education on the risk of developing a stroke.

Methods

Participants and Data Collection

In total 4434 participants were included in the study from 1956 to 1968. Participants were followed over a period of 24 years after their inclusion with follow up appointments every 6 years. For this cross-sectional analysis, a set of variables were identified and measured at baseline. For 4242 participants all variables were recorded (complete cases). For further analysis we worked only with these complete cases. In table 1 an overview is shown as provided by the Framingham Heart Study Longitudinal Data Documentation¹ of the measured variables and their distribution over the participants.

¹https://biolincc.nhlbi.nih.gov/media/teachingstudies/FHS_Teaching_Longitudinal_Data_Documentation_2021a.pdf

Variable	Description	Unit	Range or counts	Missing values
EDUC	Attained education	0: High school or lower 1: College or higher	0: 3103 1: 1218	113
AGE	Age	Years	32 - 70	0
SEX	Participant sex	1: Men 2: Women	1: 1944 2: 2490	0
BMI	Body Mass Index	Weight (kg) / Length ² (m)	15.54 - 56.8	19
DIABETES	Diabetic as casual glucose of 200 mg/dL or more	0: Not a diabetic 1: Diabetic	0: 4313 1: 121	0
CURSMOKE	Current cigarette smoking	0: Not current smoker 1: Current smoker	0: 2253 1: 2181	0
BPMEDS	Use of antihypertensive medication	0: Not currently used 1: Currently used	0: 4229 1: 144	61
HYPERTEN	Systolic blood pressure higher than 140 mmHg or diastolic blood pressure lower than 90mmHg	0: No hypertension 1: Hypertension	0: 1182 1: 3252	0
STROKE	Atherothrombotic infarction, Cerebral Embolism, Intracerebral Hemorrhage, or Subarachnoid Hemorrhage or Fatal Cerebrovascular Disease	0: No stroke 1: Stroke	0: 4019 1: 415	0

Table 1: Description and distribution of measured variables. All descriptions, units and range or counts are extracted from the Framingham Heart Study Longitudinal Data Documentation as provided by the lectures for this analysis. All variables, excluding stroke and hypertension, are from the baseline measurement.

Directed Acyclic Graph (DAG)

The assumed causal relationship from education (EDUC) to the risk of stroke (STROKE) in our DAG is modelled by a series of mediators and confounders (see figure 1). These assumptions have been made by a mixture of subjective expert opinion and literature research. First, blood pressure medication (BPMEDS), and Body Mass Index (BMI) are thought to causally affect Hypertension status (HYPERTEN), which is then causally affecting stroke risk (STROKE). Highly educated people are more likely to follow advice from medical authorities and take blood pressure medication according to its prescription, which is then in turn thought to decrease hypertension. Furthermore, highly educated people are more likely to have an active healthy lifestyle, or at least know about factors that are negatively affecting their health. This is also assumed to be true for nutrition and exercising, which is why highly educated people are assumed to have a lower BMI. A lower BMI is decreasing hypertension according to previous research. Moreover, previous research linked increased hypertension to a risk of stroke, which is also true for the effect of having diabetes (DIABETES) on stroke risk. Lastly, higher educated people are also thought to know about the negative effects of smoking for overall health. Moreover, we assume that higher education also leads to a higher income and a more attractive lifestyle, which makes the occurrence of smoking as a coping mechanism for bad life circumstances less likely. Therefore, we assume that higher education is decreasing the risk of smoking addiction. The negative causal effect of current smoking status (CURSMOKE) on the risk of stroke is well established in the literature. Besides these mediators of the causal effect, age and sex were thought to act as exogenous variables forming a backdoor path and induce potential confounding to our causal effect of interest as these variables are well-established confounders in many medical causal pathways of interest.

To control for all potential biases between education and stroke as stated by the backdoor criterion (Pearl, 2009), which is why we include the variables sex and age in the adjustment set, because these variables are acting as a backdoor path in our DAG. This allows us to analytically isolate the causal effect of education on the risk of developing a stroke in further analysis. To test the validity of the DAG with the empirical data, we tested all implied conditional independencies. For all variables (continuous and binary) we assume that by the Law of Large Numbers we can estimate the partial correlations between each set of variables.

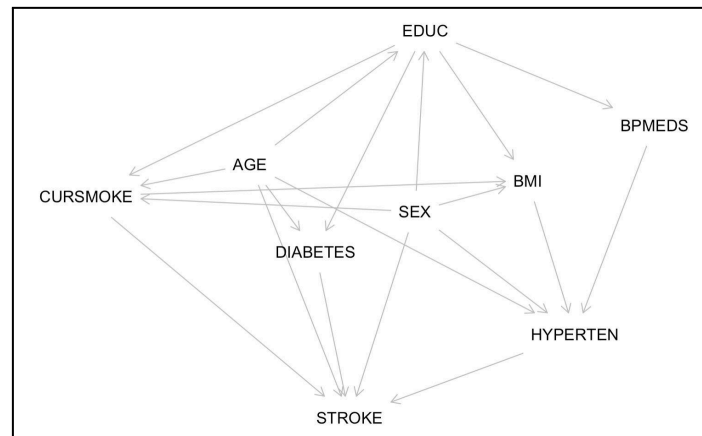


Figure 1: Directed acyclic graph (DAG) after adjustment for conditional independence. The DAG shows assumptions on how variables are associated with each other. Age and Sex meet the backdoor criterion and are included in the adjustment set further for analysis.

Propensity Score Reweighting

For estimation of the causal effect of education level on the risk of developing a stroke 24 years after inclusion in the study, we conducted Propensity Score (PS) reweighting using Inverse Probability of Treatment Weights (IPW). The PS was defined as the probability of having a high education given the measured confounders (age and sex) at baseline. The PS was estimated by fitting a logistic regression model. The IPW was calculated as one over the PS for individuals with a high education and as one over (1 - PS) for individuals with a low education. The Standardised Mean Difference (SMD) was calculated to estimate balance of the covariates before and after adjustment.

Estimating Average Causal Effect of Education on Stroke

Using the weighted data as provided by the PS weighting procedure, we continued with a logistic regression to estimate the Average Treatment Effect (ATE) of education on stroke development risk. The ATE represents the estimated difference in the probability of developing a stroke if everyone were to have had higher education, versus if everyone were to have had lower education.

Assumptions

For the analysis, the following assumptions must hold to provide valid results:

Consistency: The potential outcome as calculated by the analysis is equal to the actual observed outcome. This follows from a well defined measure for education level, and from no interference between participants.

Positivity: The probability for each individual to have a low or high education level is between 0 and 1 and there should be enough overlap between the two groups. This can be tested by visualising the overlap in PS distributions between the two education level groups.

(Conditional) exchangeability: Individuals with high education and with low education are comparable when controlled for the confounding variables. This means that all individuals would have the same potential outcome, would they all have had the same education. Furthermore, the education level is independent of

the outcome (stroke). This is tested by estimating the covariate balance before and after applying the IPWs. We used SMD, aiming for values below 0.1 after weighting.

Software

All analyses were performed using R version 4.3.3 or later. The dagitty package was used for the visualisation and analysing the DAG, the cobalt and survey packages for propensity score weighting and the Mice package was used for missing data analysis

Missing Data Analysis

Of the total of 4434 participants, 4242 complete cases were recorded. For the remaining 192 participants there was at least one measurement on a covariate missing (see table 1). We assessed the missing mechanism by investigating how the potentially confounding covariates of interest are distributed between patients for which education was observed and for which it was not. Furthermore, we ran logistic regression to investigate if any of the included covariates predicts the probability of education missing, which would act as an indicator for missing at random (MAR) or missing not at random (MNAR).

Results

Conditional Independencies of the DAG

After testing for non-zero partial correlations and applying the Bonferonni-Holm correction for multiple testing, we find three conditional independencies for which the correlations are non-zero. These are between current smoking status and BMI ($r(16) = -.17, p < .001$), current smoking status and sex ($-.20, < .001$), and current smoking status and age ($-.21, < .001$).

PS Model and IPW Adjusting

Age and education are significantly associated ($\beta = -.019, z = -4.69, p < 0.001$), whilst sex and education were not ($-.003, -0.043, .966$). The PS for low education ranged from 0.211 to 0.353, and from 0.214 to 0.349 for high education (see figure 2A).

Before adjustment, age is not well balanced between the two groups of education level ($SMD = -.163$), whilst sex is ($-.002$). After adjustment by the PS, age and sex are both well balanced ($-.006; -.001$) (see figure 2B). The effective sample size in the low education group went from 3045 to 3038 and from 1196 to 1180 in the high education group.

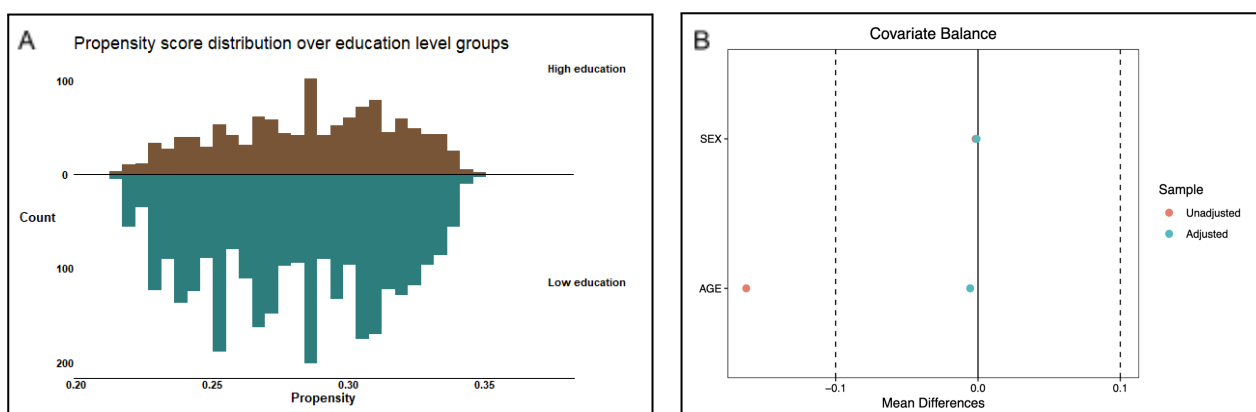


Figure 2: Propensity score distribution and Standardised Mean differences for sex and age before and after IPW-adjustment. A: PS distribution over two education level groups. The histogram shows a high overlap between propensity scores. B: Age was not well balanced before adjustment. Both sex and age are balanced over the two groups after adjustment. Dashed lines indicate the balance threshold of 0.1.

ATE Estimation

The mean probability of developing a stroke for the low education group is .0987, and for the high education group .0778. This translates to 9.87%, with 95% CI [8.89, 10.92] and 7.78% [6.47, 7.97] respectively. The

estimation for the ATE, calculated as the difference in potential stroke risk between the low education group and the high education group ($E[Y(\text{LowEd})] - E[Y(\text{HighEd})]$), was 0.021. The corresponding 95% CI was [0.002, 0.040]. Since this interval does not contain zero, the result indicates a statistically significant difference at the $\alpha=0.05$ level.

This suggests that having attained lower education is associated with a significantly higher risk of developing a stroke compared to higher education, estimated at an increase of 2.1 percentage points.

Missing Data Analysis

Logistic regression with all covariates of interest predicting the log-odds of missing showed that BMI significantly increased the probability of education being missing ($OR \approx 1.06$). A two-sample t-test showed that BMI significantly differs between participants whose education was ($M = 25.81$, $SD = 4.05$) and was not (26.90 , 4.10) observed ($t(117.01) = -2.78$, $p < 0.01$). Practically, although BMI is statistically associated with missingness in education, the effect is very small and the average BMI differs by just ~ 1 unit between complete and missing cases for education status. Both groups would still be classified as overweight according to BMI classification criteria ($BMI > 25$), and given the BMI measurement range this difference is practically negligible. Therefore, while this technically violates MCAR, the violation is minor and likely not practically important. Furthermore, since all other covariates were almost identically distributed between groups, we do not assume external unobserved covariates influencing the missingness mechanism. For many purposes, the data may be treated as MCAR or MAR with minimal bias risk given the empirical evidence collected, although MNAR can not be ruled out completely.

Conclusion

This cross-sectional analysis aimed at finding the causal effect of education on the risk for developing a stroke. By controlling for the confounding factors of sex and age using propensity score weighting, we isolated the estimated causal effect of education. We found a statistically significant result, estimating that higher education is associated with a 2.1 percentage point lower risk of developing stroke over 24 years compared to lower education ($ATE = 0.021$, 95% CI [0.002, 0.040], representing the estimated risk difference $E[Y(\text{LowEd})] - E[Y(\text{HighEd})]$). Therefore, our finding supports the hypothesis that higher educational attainment has a protective causal effect against stroke, independent of baseline age and sex. While the magnitude of the risk reduction is modest, its statistical significance provides supporting evidence that a causal effect exists after accounting for the confounders.

Despite this finding, several limitations should be acknowledged. Due to time constraints we decided not to adapt our complete case analysis to an imputation-based analysis. The data shows no indication for a Missing not at Random mechanism for the missing data, meaning that the missing values can be considered random. Therefore we could have estimated the missing values using multiple imputation. For further research we do suggest using multiple imputation for a more robust estimation.

Our second suggestion is to examine the underlying distribution of our considered variables. In the missing data analysis it became clear that for the two groups where participants did and did not report on their education level, the mean BMI was considered to be overweight ($BMI > 25$). Together with this, we also saw low frequencies of crucial mediators in combinations.

All in all, the current study supports original findings on the association between education and the risk of developing a stroke in the long term by providing statistically significant evidence for a causal link after adjusting for confounding. We isolated the causal effect of education and thereby estimated a direct measure for the risk difference. While this study strengthens the evidence for a protective effect, how education influences stroke development remains unclear. For developing effective interventions we consider it to be insightful to look into this underlying mechanism.

All authors contributed equally to the report.

Please find the additional R-code in the separate RMD-file.

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